BIO-DATA / CURRICULUM VITAE

DR. ANIKA SHARMA

Scientist E — Atmospheric Instrumentation & Applied AI

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1. PERSONAL AND OFFICIAL PARTICULARS

Name	Anika Sharma
Present designation	Scientist E
Present organisation	Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory
Present pay level	Level 12 (7th CPC)
Email	anika.sharma@example.org
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2. EDUCATIONAL QUALIFICATIONS

Qualification	Discipline / Subject	Institution / University	Year	Division / Grade
B.E.	Electronics and Communication Engineering	Northfield Institute of Science and Technology, Northfield University	2004	First Class with Distinction
M.Tech.	Avionics Thesis: Fault-tolerant data acquisition for airborne test instrumentation	Coastal Institute of Aeronautical Engineering	2010	CGPA 8.6/10
Ph.D.	Electrical and Electronics Engineering Thesis: Lightweight deep networks for real-time enhancement of aerial imagery	Northfield Institute of Science and Technology	2021 – ongoing	Ongoing

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3. DETAILS OF EMPLOYMENT (IN CHRONOLOGICAL ORDER)

Office / Organisation	Post held	From	To	Pay level	Nature of duties
Test Systems Group, Meridian Electronics Limited	Graduate Engineer	08/2002	05/2004		Built automated test equipment for avionics line-replaceable units.
Avionics Integration Group, Halcyon Avionics Design Bureau	Senior Design Engineer	06/2004	06/2016		Design, integration and environmental qualification of avionics line-replaceable units; development of automated test rigs; support to flight trials of UAV payloads.
Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory	Scientist E	07/2016	Till date	Level 12	Planning, testing and maintenance of radiosonde ground systems and upper-air instruments; management of the division workshop; leadership of IoT monitoring and AI/ML development; technical evaluation and procurement of instruments under GFR; guidance of project staff and interns. Additional charge: Workshop In-charge, Upper Air Division Workshop (04/2019 – Present) Nodal officer: Nodal Officer, IoT and AI/ML Initiatives (01/2021 – Present)

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4. QUALIFICATIONS AND EXPERIENCE AGAINST THE REQUIREMENTS OF THE POST

S. No.	Qualifications / experience required (as advertised)	Qualifications / experience possessed
1	Essential: Master's degree in Electronics / Electronics and Communication / Avionics Engineering from a recognised university.	M.Tech., Avionics — Coastal Institute of Aeronautical Engineering (2010)
2	Essential: At least 10 years of experience in design, integration or testing of avionics, embedded or sensor systems.	At least 23 years (since Aug 2002) in avionics, test & evaluation: Scientist E, Kestrel Atmospheric Instrumentation Laboratory; Senior Design Engineer, Halcyon Avionics Design Bureau; Graduate Engineer, Meridian Electronics Limited
3	Desirable: Ph.D. in Electronics, Electrical or Computer Engineering.	Ph.D., Electrical and Electronics Engineering — Northfield Institute of Science and Technology (ongoing)
4	Desirable: Experience in machine learning, computer vision or edge AI deployment for airborne or UAV platforms.	Object detection (skill); Image super-resolution (skill); Edge super-resolution pipeline for UAV imagery (project)
5	Desirable: Publications in peer-reviewed journals.	6 peer-reviewed publication(s): Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs; An IoT telemetry framework for upper-air ground stations; A low-cost GNSS receiver chain for radiosonde tracking
6	Desirable: Experience of leading technical teams and project coordination.	Radiosonde ground-station acceptance test protocol (project); Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for b (achievement / responsibility)
7	Desirable: Design and evaluation of airborne sensor and avionics subsystems	Mentored graduate engineer trainees on avionics bus protocols and test documentation. (achievement / responsibility); Developed LabVIEW test sequences for power-supply and interface boards of avionics units. (achievement / responsibility); IoT monitoring of laboratory infrastructure (project)
8	Desirable: development of embedded processing for UAV payloads	Edge super-resolution pipeline for UAV imagery (project); Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded de (achievement / responsibility); Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight (achievement / responsibility)
9	Desirable: preparation of technical specifications and test plans	Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Governm (achievement / responsibility); Technical procurement (GeM, GFR) (skill)
10	Desirable: guidance of junior scientists.	A low-cost GNSS receiver chain for radiosonde tracking (publication); Scientist E (position)

5. PUBLICATIONS

[1] A. Sharma and R. Iyer, “Knowledge-distilled lightweight networks for drone image super-resolution,” accepted for publication in Example Transactions on Aerial Imaging.

[2] A. Sharma, K. Rao and R. Iyer, “Benchmarking embedded GPUs for real-time image enhancement,” Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] A. Sharma and R. Iyer, “Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs,” Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

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[4] A. Sharma, P. Rao and S. Gill, “An IoT telemetry framework for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] A. Sharma and S. Verma, “Edge telemetry dashboards for upper-air ground stations,” Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] A. Sharma, P. Menon and R. Iyer, “A low-cost GNSS receiver chain for radiosonde tracking,” Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

6. MANUSCRIPTS UNDER REVIEW / IN PREPARATION

[1] A. Sharma, M. Khan and R. Iyer, “Detection-aware quality assessment of super-resolved drone images,” under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] A. Sharma, “Simulation study of balloon-released gliding sondes,” in preparation. *[In preparation]*

7. MAJOR PROJECTS AND TECHNICAL CONTRIBUTIONS

Edge super-resolution pipeline for UAV imagery *(Ongoing)* 09/2021 – Present

Principal investigator, Kestrel Atmospheric Instrumentation Laboratory

Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.

Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Radiosonde ground-station acceptance test protocol 01/2018 – 03/2019

Coordinator

Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

IoT monitoring of laboratory infrastructure 02/2020 – 06/2022

Project lead

Instrument rooms lacked continuous environmental and power-quality monitoring.

Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

8. TRAINING PROGRAMMES AND CERTIFICATIONS

Deep Learning for Edge Devices (40-hour course) 2022

Example Academy of Applied Computing

Public Procurement and GFR 2017 (5-day programme) 2019

Example Institute of Public Administration

9. AWARDS & HONOURS

Best Paper Award 2023

Example Conference on Instrumentation (ECI 2023)

10. ADDITIONAL RESPONSIBILITIES AND COMMITTEES

Supervisor, M.Tech. interns in applied machine learning 2022 – Present

Kestrel Atmospheric Instrumentation Laboratory

Supervise summer and dissertation interns on edge-AI projects.

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Chair, Technical Evaluation Committee for radiosonde procurement 04/2023 – 02/2024
Kestrel Atmospheric Instrumentation Laboratory
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

11. DECLARATION

I hereby declare that the particulars furnished above are true, complete and correct to the best of my knowledge and belief.

Place: New Delhi
Date: (Signature)
Anika Sharma

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